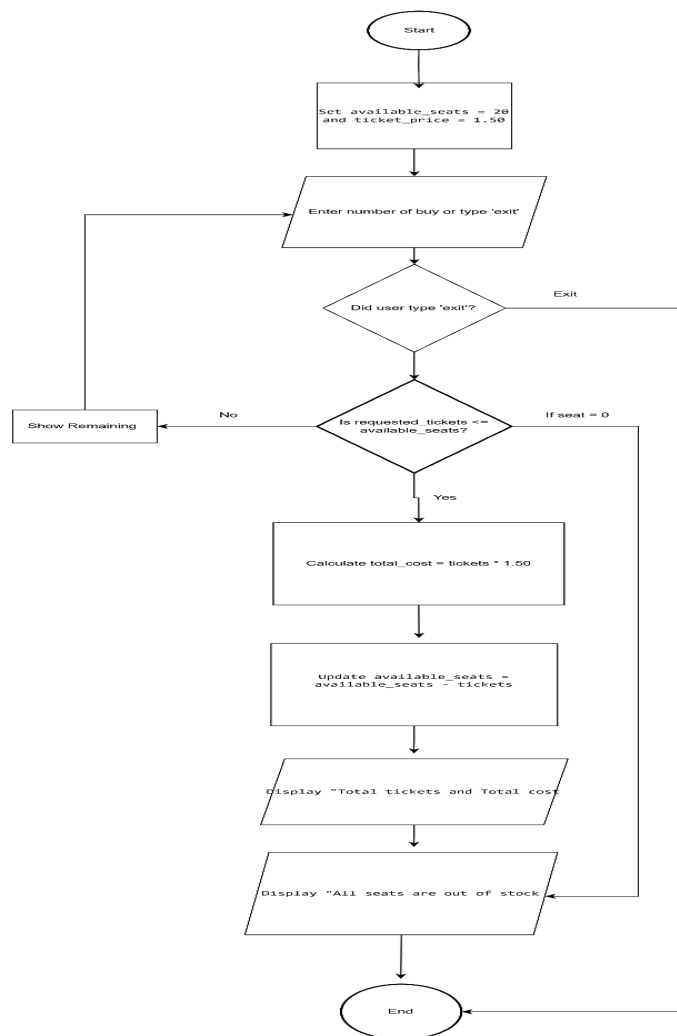


Green Campus Shuttle Ticket Kiosk - Assignment

Task 1: Algorithm Design (Logic Planning)

1. **Set Up:** Start the program and set the seats to 20 and the price to \$1.50.
2. **Start Loop:** Use a loop so the program keeps asking for the next student until the bus is full or you stop it.
3. **Ask Operator:** Ask the person using the computer if they want to "buy" a ticket or "exit" the system.
4. **Check for Exit:** If they type "exit," close the program right away.
5. **Ask for Tickets:** If they want to buy, ask them "How many tickets do you want to buy?".
6. **Input Safety:** Make sure the program doesn't crash if they accidentally type letters instead of a number.
7. **Check Seats:** Look to see if there are enough seats left for the tickets they want.
8. **Error Message:** If they ask for too many tickets, print an error saying there isn't enough room and show how many seats are left.
9. **Do the Math:** If seats are available, multiply the number of tickets by \$1.50 and subtract the tickets from the total seats.
10. **Finish Up:** Show the total cost and remaining seats on the screen. If the seats hit 0, tell them the bus is full and stop the program.

Task 2: Logic Flowchart (Visual Representation)



Task 3: Implementation (Python Code)

#Set up

available_seats = 20

ticket_price = 1.50

system_active = True

print("--- Welcome to Green Campus Shuttle Ticket Kiosk ---")

while system_active:

#Input/Output - Get user intent

choice = input("\nWould you like to buy a ticket or exit? (buy/exit): ").lower()

if choice == "exit":

print("Closing system. Have a great day!")

system_active = False

elif choice == "buy":

Get ticket quantity from user

try:

requested_tickets = int(input(f"How many tickets? (Seats left: {available_seats}): "))

Check if enough seats remain

if requested_tickets > available_seats:

print(f"Error: Not enough seats! Only {available_seats} remaining.")

elif requested_tickets <= 0:

```
print("Please enter a valid number of tickets.")
```

```
else:
```

```
    # Calculate cost and update seats
```

```
    total_cost = requested_tickets * ticket_price
```

```
    available_seats -= requested_tickets # Updates available_seats
```

```
    # Display Receipt
```

```
    print("-" * 30)
```

```
    print(f"RECEIPT:           {requested_tickets} Tickets")
```

```
    print(f"TOTAL COST:       ${total_cost:.2f}")
```

```
    print(f"REMAINING SEATS:    {available_seats}")
```

```
    print("-" * 30)
```

```
except ValueError:
```

```
    print("Invalid input! Please enter a whole number for tickets.")
```

```
#Auto-close if bus is full
```

```
if available_seats == 0:
```

```
    print("\nAll seats are now full!")
```

```
    system_active = False
```